

Henry Noyes

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EDUCATION

Northeastern University

Boston, MA

MS in Robotics | GPA: 4.00

August 2026

BS in Mathematics and Physics, Minor in Data Science | GPA: 4.00

May 2024

Honors: NASA Space Technology Graduate Research Fellow, 2022 NASA BIG Idea Challenge Artemis Award, Massachusetts Space Grant Consortium Fellowship, ASME SMRDC 1st Place, Dean's List, National Merit Scholarship

Relevant Coursework: Reinforcement Learning, Control Systems Engineering, Practical Neural Networks, Mobile Robotics, Multimessenger Astrophysics, Machine Learning, Classical Dynamics, Group Theory

RESEARCH EXPERIENCE

Silicon Synapse Lab, Northeastern University

Boston, MA

Student Researcher

Jan 2022–Aug 2026

- Co-created the COBRA system (Crater Observing Bio-inspired Rolling Articulator), a modular snake robot capable of sidewinding and tumbling with a mission to measure ice concentration in permanently shadowed lunar craters
- Selected as 1 of 60 national recipients of the 2024 NASA Space Technology Graduate Research Opportunities (NSTGRO) fellowship to advance autonomy in snake robots for lunar exploration
- Served as electrical lead for a team of 11 students selected as 1 of 7 finalists for the 2022 NASA BIG Idea Challenge, where we won the Artemis Award—the top honor of the competition
- Led the lab-wide adoption of modern robotic simulators for reinforcement learning research, creating the open-source tools [robot-see](#) and [robot-diff](#) to assist in model development
- Developed and deployed a hierarchical control architecture for autonomous whole-body loco-manipulation with a snake robot, demonstrating robust non-prehensile transport of a target object to arbitrary goal locations in a motion capture arena

Robotic Surface Mobility Group, NASA Jet Propulsion Laboratory

Pasadena, CA

Visiting Graduate Researcher

June–Aug 2026

- Trained control policies for snake robot locomotion with reinforcement learning in MuJoCo and investigated the impact of contact observations on task performance
- Deployed and evaluated learned policies onboard a snake robot with custom force-sensing modules

Robotics and Intelligent Vehicles Research Lab, Northeastern University

Boston, MA

Robotics Engineer

Jan–June 2023

- Led the data processing and system integration efforts on a DARPA-funded mixed reality project focused on perceptually-enabled task guidance (PTG)
- Developed a ROS2 architecture to efficiently collect and store egocentric data from the Microsoft HoloLens 2 headset
- Assisted in training object detection and action recognition models for deployment on an augmented reality headset, and created an automated object annotation pipeline using OptiTrack motion capture

SKILLS

Languages: Python, C++, MATLAB, Bash, LaTeX

Frameworks/Applications: Isaac Sim/Lab, MuJoCo, ROS/ROS2, Git, Docker, SolidWorks, Blender

Hardware: 3D printing (FDM, SLA), soldering, laser cutting

PUBLICATIONS

A. Salagame, **H. Noyes**, H. Hublikar, and A. Ramezani, “Autonomous Whole-body Loco-manipulation Using a Snake Robot,” *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026

A. Salagame, **H. Noyes**, E. Sihite, A. Kalantari, and A. Ramezani, “Crater Observing Bioinspired Rolling Articulator (COBRA),” *Advanced Intelligent Systems*, 2025, <https://doi.org/10.1002/aisy.202500352>